



Taking notes with artificial intelligence (AI)-based technology and academic integrity: perspectives from online learners

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Abstract

This study examines the intersection of online graduate students' note-taking practices, technology use, and academic integrity, emphasizing the impact of artificial intelligence (AI) tools. Using a post-phenomenological case study approach, multiple in-depth interviews were conducted with eight graduate learners from a Canadian university, gathering qualitative insights into diverse perspectives on how note-taking technologies mediate their understanding of learning and ethical behaviour. Findings reveal that AI-powered tools, such as automated note-takers, influence learners' note-taking strategies by prioritizing efficiency and accessibility, often at the expense of critical engagement. While academic integrity remains a key concern in formal assignments, ethical considerations are largely absent in informal contexts, where practices like copying content or using AI-generated summaries are normalized. Learners' preferences lean toward usability and off-camera learning, with minimal emphasis on the ethical implications of technology use. These findings highlight the growing need for educational support and clearer guidelines on integrating AI ethically into academic practices, ensuring that these tools reinforce critical thinking and accountability while adapting to evolving educational landscapes.

Keywords Note-taking · Academic integrity · Educational technologies · Online learning · Postphenomenology · Artificial Intelligence (AI)

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1 Introduction

The integration of artificial intelligence (AI) technologies into online learning has fundamentally transformed how higher education students engage with and process information. AI-powered tools, such as automated note-taking applications, summarize lectures, readings, and discussions in real-time, enabling students to focus on comprehension rather than transcription. While these tools enhance accessibility and efficiency, they also reshape students' perceptions of academic integrity by blurring the boundaries between legitimate academic assistance and academic dishonesty. For instance, when AI-generated summaries replace original, student-created notes, learners may develop a diminished sense of ownership over their academic work. This shift is significant because it challenges traditional definitions of intellectual effort, potentially fostering a culture of over-reliance on automation. As AI continues to evolve, educators and institutions must critically address these implications, ensuring that AI is leveraged ethically to support—not undermine—the development of critical thinking and accountability in online learning environments. By examining these changes, we can better prepare for an educational landscape that balances technological innovation with the preservation of core academic values. Using a post-phenomenological approach, this paper examines how online learners are currently negotiating these challenges, adapting their note-taking strategies, and interpreting academic integrity in higher education.

2 Literature review

Artificial intelligence (AI) has transformed educational landscapes, particularly through tools designed to augment student learning processes. The rapid evolution of AI-powered tools, including automated note-taking applications, has offered ways to streamline the learning experience by capturing and organizing information efficiently, while also raising questions around academic integrity, learner autonomy, and the implications for traditional pedagogical practices (e.g., Cotton et al., 2024). The following section will focus on AI use in the note-taking process as a key step for learners to consolidate knowledge.

2.1 Note-taking for online learners

Note-taking is a foundational strategy for learning and knowledge retention, used in both academic and non-academic settings to reinforce understanding and aid recall. Academic literature identifies multiple methods for note-taking, including traditional pen-and-paper techniques, digital notes, and hybrid approaches, with learners often tailoring their methods based on individual preferences, study context, and learning objectives (Chudaeva, 2024). Research consistently demonstrates that effective note-taking supports cognitive processes like encoding and external information storage, while also facilitating the organization of main ideas and enhancing reflection (Boch & Piolat, 2005; Crooks et al., 2007; Fisher & Harris, 1973; Hadwin et al., 1999; Kirby et al., 1999; Nakayama et al., 2014; Shier, 2020). Yet, despite these benefits,

Morehead et al. (2019a, b) found that online learners often take fewer notes than those in traditional settings, suggesting a need for more accessible digital tools and strategies that encourage metacognition development in online learners.

As online learning typically requires a high degree of self-directedness, effective note-taking is integral to academic success in this setting. Research has shown note-taking to be a metacognitive, self-monitoring process that is crucial for self-regulated learning (Kauffman et al., 2011), and online learners who consistently take notes tend to perform better academically (Filcher & Miller, 2000). Hence, note-taking is both a practical and cognitive tool that enhances engagement and learning outcomes in online education.

2.2 Tools and technologies for note-taking

Learners use various tools for learning and note-taking to support their learning. Some online learners use pen and paper, while others leverage digital tools or devices like laptops and apps for taking notes. Previous research explored note-taking technologies such as iPads (for example, Pyorala et al., 2019), cloud-based tools like Microsoft OneNote (2016) and Google Docs (2024) (Gombas, 2020), mobile devices (Mfaume et al., 2018; van Wyk & van Ryneveld, 2018), collaborative note-taking applications (Petko et al., 2019; Reilly & Shen, 2011), digital pen use (Shell et al., 2021), and longhand note-taking (Morehead et al., 2019a; Mueller & Oppenheimer, 2014).

With the proliferation of AI enhanced features, note-taking tools such as Otter.ai (<https://otter.ai/>) and Notion (<https://www.notion.so/>) have begun leveraging natural language processing models to support learners in transcribing and summarizing lecture note-taking in real-time. These AI note-taking functionalities enable learners to focus on understanding the materials rather than simply capturing it (Kim et al., 2009). Once saved, the notes can then be further organized, analyzed, shared, and are formatted in a searchable file to streamline study processes and retention (Johnson & Smith, 2022). As a result, note-taking technologies have also become more inclusive and accessible (e.g., Boyle & Joyce, 2019), enabling learners with diverse learning needs to readily employ speech-to-text capabilities, closed captioning, magnification, and other personalized features. While AI can provide tailored resources and feedback to support learner engagement, these technologies also provide opportunities for empowering learners to take pride, ownership and develop authentic progress in their learning. Some studies have even shown a correlation between the use of AI note-taking tools on reduced study time and improved academic outcomes (e.g., Ward et al., 2024).

An optimal note-taking tool depends on the nature of the learning activity, the goal of the learning activity, whether notes are reviewed, and the learner's level of comfort with the technology tool(s) for taking notes (Chudaeva, 2024; Luo et al., 2018). There is evidence that handwritten and digital notes may be equally effective in terms of affecting learners' academic performance (Voyer et al., 2022). Online learners, in particular, -alternate between taking paper notes and digital notes, or take both— combining formats in a variety of ways to meet their individual needs (Veletsianos, 2020).

2.3 Disruptive generative artificial intelligence (GenAI) tools on academic integrity and educational policy

The emergence of more sophisticated Generative AI (GenAI) tools has become a transformative force in higher education, bringing its own set of challenges and opportunities for educators and learners (Dai et al., 2023; Nartey, 2024; Wang et al., 2024). While GenAI facilitates accessibility, inclusion, and personalized learning experiences in some ways, it also heightens existing issues related to academic integrity, privacy, and security. These tensions invite us to examine the “technological intentionality” (Irwin, 2016) embedded in GenAI tools, which shape how they are perceived and used in learning contexts.

Take for instance, AI-powered note-taking software like Otter.ai (<https://otter.ai/>) and Notion (<https://www.notion.so/>) — these GenAI tools are reshaping how learners capture, organize, and recall information. Specifically, Otter.ai introduced an automated “note-taking participant” feature that autonomously joins calendar meetings, mimicking human-like intentions and capabilities. While this feature may improve convenience, it also raises ethical, ecological, and existential questions about what it means to “engage and learn” alongside AI companions. From a postphenomenological standpoint, technology embodies intentionality that modifies users’ experiences — enhancing some perceptual or actionable aspects (e.g., the speed and accessibility of note-taking) while diminishing others (e.g., cognitive engagement with the material). Traditional pen-and-paper note-taking, for instance, aligns with a slower, sequential process that requires deliberate effort, which can promote reflective and selective engagement. Past research suggests that the act of writing by hand fosters deeper cognitive processing and retention, as users actively synthesize information (e.g., Mueller & Oppenheimer, 2014). This intentional, manual approach can enhance comprehension and memory, as the constraints of pen and paper necessitate focus and intentionality in a way that typing on a computer keyboard does not.

Rather, digital note-taking tools such as Microsoft OneNote (2016) bring about a different form of technological intentionality. Designed for rapid and flexible note capture, these tools support multimedia-rich entries with images, audio, tags, and links, enabling ease of organization and retrieval. While digital notes improve efficiency, the speed and ease with which users can record information may reduce engagement or cognitive encoding during the initial learning process, as users may capture material passively rather than processing it deeply. The flexibility of editing and reordering digital content can also result in a more transient and less focused approach to information capture, prioritizing recall over retention. The ease of use in AI note-taking tools also raises concerns about academic integrity. While structured documentation in digital notes may help learners avoid unintentional plagiarism, the ability to copy and paste from AI-generated notes presents potential for misuse. Although AI can support citation management and content generation, research indicates that it may also enable academic dishonesty—prompting calls for stronger policies and ethical education to guide its use (Mumtaz et al., 2024). Thus, educators and institutions need to promote ethical usage and reinforce academic integrity policies surrounding the uptake of these technologies for learning.

Moreover, the convenience of AI tools may itself obscure ethical reflection. Learners often prioritize ease and efficiency, a tendency shaped by effort-reduction and status quo biases, which can discourage critical scrutiny of how such tools affect authorship, accountability, and learning depth. As Eaton (2021) notes in her work on *everyday ethics*, routine educational decisions—such as relying on AI-generated notes—may seem inconsequential but are in fact deeply ethical, as they shape habits, values, and the integrity of the learning process.

Universities commonly incorporate traditional note-taking methods, such as Cornell Notes (Pauk & Owens, 2010), within learner toolkits. However, to work effectively with GenAI, learners also need to develop AI literacy. This includes learning to craft effective prompts, critically evaluate outputs, and use AI responsibly in academic contexts (Acosta-Enriquez et al., 2024; Bin-Nashwan et al., 2023; Dai et al., 2023). Unlike handwritten or digital notes that can be kept private, interactions with GenAI tools are often used to train other AI models (e.g., commodified AI tools, AI detection tools), adding further privacy implications concerning how input data will be stored, applied or shared. As such, institutions such as the University of Waterloo, have pushed for greater AI literacy professional development (King 2024; Hervieux & Wheatley, 2020), emphasizing the need for learners to develop competencies on “how to learn with AI” as part of their academic growth.

2.4 Navigating AI ethics and competencies for online educators and learners

As AI tools become integral to academic practices, they raise complex ethical questions, especially around authorship, intellectual property, and the balance between support and over-reliance. Addressing these challenges underscores the need for focused professional development in AI literacy for both educators and learners to foster responsible use of these technologies while maintaining academic integrity.

2.4.1 Authorship

The traditional concept of authorship is rooted in original intellectual contribution. However, AI tools that generate text or synthesize research complicate this idea. Chan (2023) found that post-secondary learners frequently use AI to draft or summarize academic work, which blurs the line between the student’s original thoughts and those suggested or generated by AI. This creates ambiguity regarding the true origin of ideas and can dilute the authenticity of academic contributions. Clear guidelines on AI’s role in authorship are crucial to ensure that the work credited to an author truly reflects their intellectual input. Such guidelines would be strengthened by professional development in AI literacy, equipping both students and faculty with an understanding of ethical authorship practices in an AI-augmented academic landscape.

2.4.2 Intellectual property

AI tools that manage and store academic data also present issues regarding intellectual property (IP). Many of these tools handle proprietary or confidential information, necessitating policies to protect creators’ rights while leveraging AI’s efficiency. For

instance, asking learners about their expectations and ethical implications surrounding how the work they feed into AI chatbots could be re-purposed or commodified in unexpected ways (e.g., used to train AI detection software, thereby flagging their own work as AI-generated or plagiarized). As noted by the World Intellectual Property Organization (WIPO, [n.d.](#)), integrating AI in managing intellectual property calls for robust frameworks to safeguard original content and ensure compliance. Educational institutions must develop frameworks to manage IP rights responsibly, aligning them with the ethical use of AI. Through AI literacy programs, educators and students can become more aware of IP complexities, fostering a culture that respects and understands the rights of content creators.

2.4.3 Pedagogical considerations

Previous studies have shown that excessive reliance on AI for tasks such as note-taking can hinder critical thinking and diminish learners' ability to engage deeply with material. For instance, Gerlich ([2025](#)) found that while AI-supported research tools reduced learners' cognitive load and simplified the process of locating information, they also weakened students' reasoning and deep engagement typically fostered through manual information-seeking and critical evaluation. Learners who depended heavily on AI for note-taking and research demonstrated noticeably weaker independent critical analysis and synthesis skills. Educator perspectives on generative AI further illuminate these concerns. Echave et al. ([2024](#)) reported that while educators largely welcomed AI's potential, they remained apprehensive about its effects on students' cognitive development and the authenticity of learning processes. These concerns are echoed in Zhang et al. ([2024](#)), who found that instructors in Chinese universities, although experimenting with AI-assisted assessment, emphasized the need for clearer pedagogical guidelines to safeguard academic integrity and meaningful student participation. Similarly, Chukhno ([2024](#)) observed that in foreign language education, both teachers and students acknowledged AI's usefulness for personalized support but warned against its overuse, which may discourage active learning and compromise learner autonomy.

Additionally, AI note-taking technologies are not infallible. Shafiee Rad, et al. ([2023](#)) identified issues with transcription accuracy, particularly in contexts involving background noise, domain-specific terminology, or multiple speakers, which can result in incomplete or misleading notes. Moreover, AI-generated summaries, while convenient, may encourage surface-level processing and passive engagement, ultimately undermining retention, conceptual understanding, and the quality of participation in course discussions. Without mechanisms for verifying the accuracy or credibility of AI-generated content, learners may unintentionally absorb misinformation or reinforce algorithmic biases—issues that Zhai et al. ([2024](#)) emphasize must be addressed through responsible AI design. Privacy and data security risks further complicate AI's integration into educational contexts, as breaches of academic or personal data can lead to unauthorized access, reputational harm, or academic misconduct.

To harness AI's potential while maintaining academic integrity and learner engagement, institutions must proactively address these pedagogical and ethical challenges. This includes offering sustained professional development to foster AI literacy

among both students and educators, ensuring that users understand not only how to use these tools effectively, but also the broader implications related to intellectual property, bias, and ethical scholarship. Clear, institution-wide guidelines informed by emerging research and educator experiences (Echave et al., 2024; Zhang et al., 2024; Chukhno, 2024) are essential for ensuring that generative AI supports—rather than undermines—the goals of higher education.

2.5 Generative AI and Co-Authoring tools

Generative AI technologies, such as ChatGPT (<https://openai.com/api>) and Verb (<https://verb.ai>), provide co-authoring support by generating text with minimal user input, presenting both valuable educational tools and potential risks. Generative AI's capabilities in drafting text have been met with concern in academic literature, highlighting issues related to authorship, autonomy, and control (Biermann, 2022).

A recent study by Chan and Hu (2023) identified several reasons learners use GenAI, including:

- Personalized and immediate learning support.
- Writing and brainstorming assistance.
- Research and analytical support.
- Multi-modal content creation, such as visual and audio support.
- Administrative tasks and organization.

One study participant noted using ChatGPT to explore class topics and receive new ideas—examples of *cognitive offloading*, which refers to using technology to reduce the cognitive demands of a task (Risko & Gilbert, 2016). While these activities may not constitute academic misconduct if properly cited, they raise questions about the ethical boundaries of AI in academic work (Perkins, 2023).

In navigating these boundaries, it becomes essential for institutions to clarify acceptable uses of AI tools, balancing innovation with academic integrity. Addressing this complexity requires nuanced academic integrity policies that acknowledge the varied ways learners employ AI in their learning processes, recognizing both ethical and academic impacts (Birks & Clare, 2023). For example, AI use in higher education courses depends largely on the learning objectives, academic integrity policies and context outlined by the course instructor. Within our research study, one participant specifically clarified that they would never use the output of ChatGPT in their assignments because AI trust, or explainability, is still a topmost concern. They felt the AI tool served as a helpful brainstorming partner to seek out additional research leads, but might be detrimental to their writing skill development in the long-term.

As educators, we need to consider a range of possible ways that learners may use AI tools. Overall, it is a challenge to make ethical judgments about learner usage of such technologies (Perkins, 2023) and in particular, how to authentically assess and compare work when AI is involved during the conceptualization or writing stages. However, we could say that it is not academic misconduct if how these tools had been used were properly cited by learners in their academic work.

2.6 The role of academic integrity on educational standards and policy

In higher education, academic integrity is increasingly recognized as integral to establishing institutional quality standards (Glendinning, 2022). AI tools pose both opportunities and risks to these standards, as they are rapidly becoming accessible and widely used in academia, often in ways that emphasize learner's exploration over institutional knowledge or instructional design practices (e.g., Yuen, 2024; Crompton & Burke, 2023). As AI-powered tools are primarily commercial products, they risk being detached from not-for-profit educational purposes, raising ethical concerns around accessibility, transparency, and the commodification of educational technologies (Memarian & Doleck, 2024).

There is also a pressing need to explore the broader implications of AI in teaching and learning, including its potential to reshape instructional practices, data ethics, and the social responsibility of educational institutions (Glendinning, 2022). One approach to addressing these complexities is by re-conceptualizing academic integrity as extending beyond learner conduct—towards the integration of broader institutional and professional ethics (Eaton et al., 2023). This holistic approach, called the *Comprehensive Academic Integrity Framework*, posits that academic integrity forms a foundational pillar across educational contexts, beyond mere learner conduct. This framework identifies eight interrelated domains of ethical practice (Eaton, 2024):

1. Everyday Ethics: Promoting ethical behavior in daily interactions.
2. Institutional Ethics: Embedding integrity within organizational policies and practices.
3. Ethical Leadership: Encouraging ethical decision-making by educational leaders.
4. Professional and Collegial Ethics: Ensuring high ethical standards among academic peers.
5. Instructional Ethics: Maintaining integrity in teaching and assessment practices.
6. Learner Academic Conduct: Addressing and guiding ethical behavior in learners.
7. Research Integrity and Ethics: Upholding ethical standards in research practices.

Eaton's framework reflects a multi-dimensional view of academic integrity, aligning it with educational systems rather than to a singular, learner-centered concern. Institutions, therefore, are encouraged to implement academic integrity as a core component of quality assurance and policy, ensuring alignment across all levels of academic and administrative conduct (Eaton, 2023).

3 Research design

To understand the relationships between note-taking technologies and academic integrity behaviour, as well as to explore online graduate learners' perceptions and experiences with note-taking practices, our study probed the following questions:

- (1) What factors affect learners' decision-making and/or selection of technologies for note-taking?

- (2) What are learners' perspectives on academic integrity when taking notes for learning?

3.1 Participants

This study consists of multiple-stage, in-depth interviews with $n=8$ graduate learners at an online Canadian university. Seven participants were domestic learners and one is an international learner. Due to participant availability and volunteer sampling constraints, all participants were female between 30 and 69 years old. This convenience sample therefore presents a limitation in terms of generalizing the findings to a wider population. The data was collected between Winter 2023 to Spring 2024 when participants were in their second half or close to the end of the program. Data collection tools included virtual interviews via Microsoft Teams. Participants range from 3 to 13 years of post-secondary education experience. Many learners have been taking one course per semester in their graduate online studies due to family and work commitments and might have had some breaks in between courses or programs of study. Participants have as little as two years of experience in an online program and as many as eight years taking courses online (Table 1).

All participants reported a strong understanding of the requirements of an online learning environment, including its challenges and the technology tools and skills necessary for successful and meaningful participation. Each participant had access to at least one personal electronic device, with many owning multiple devices, which they used to access online resources and complete assignments. Specifically, participants reported using laptops, desktops, or iPads for school-related activities, with none relying on smartphones for regular academic tasks.

3.2 Methodology

To address our research questions, we adopted a qualitative research paradigm (Lincoln & Guba, 1985) and a postphenomenological case study design focused on exploring human-technology-world relationships (Irwin, 2016). Postphenomenology, the guiding methodology, recognizes that technologies are not neutral intermediaries between humans and the world, but rather active mediators that shape our perceptions and actions. As Verbeek (2005) explains, technologies “are not neutral ‘intermediaries’ between human and world, but mediators; they actively mediate this relation” (p. 114). In this postphenomenological case study, we examined note-taking technologies as they mediate the user's experience and impact choices in learning contexts. By focusing on a single type of technology, this approach allowed us to investigate how specific tools display multistability, meaning they afford a range of uses and adapt to varied user contexts. This framework also highlights the “amplification-reduction” effect of AI tools. While AI note-taking technologies amplify

Table 1 Online learning experience of participants (Original work by Chudaeva, 2024)

Number of years	Number of participants
0 to 3	4
3 to 5	3
5 +	1

some aspects of accessibility and efficiency, they may reduce opportunities for critical reflection and personalization in note-taking. The tension between these effects underscores the need for a nuanced understanding of how AI tools mediate educational experiences and reshape traditional pedagogical values (Verbeek, 2005). The postphenomenological approach also enables us to analyze the ethical dimensions and implications of these technologies in mediating human-technology interactions.

Data was generated through a combination of methods: initial and follow-up interviews, a note-taking activity paired with self-reflection (in which participants used both analog and digital methods to reflect on their experiences), and artifact analysis of the notes produced during this activity. To minimize participant burden, they were asked to take notes on their current course material. This approach provided insights into participants' lived experiences with note-taking technologies while maintaining a naturalistic context for data generation.

Initial interview questions (see Appendix 1) were designed to explore participants' experiences with note-taking and to identify the contexts or situations that typically influenced their engagement with this phenomenon. Follow-up interviews were conducted to clarify the nature of the note-taking phenomenon and to further examine the preliminary analysis of data collected during the initial interviews, reflective exercises, and images of participants' notes. These follow-up interviews included an image elicitation component to better capture participants' perceptions of their own notes. Additionally, self-reflection (see Appendix 2) was employed as a self-reporting technique that enabled participants to articulate their perceptions, beliefs, and thoughts. This approach provided deeper insight into participants' experiences and enhanced understanding of the context surrounding their responses (Creswell & Poth, 2018; Fetsco & McClure, 2005).

Data analysis incorporated both thematic analysis and postphenomenological techniques. Thematic analysis involved coding the data, organizing it into categories, identifying recurring themes, and developing interpretations based on emergent patterns. In addition, postphenomenological techniques—specifically variational analysis—were employed to examine the affordances of note-taking technologies and their influence on participants' decision-making processes. Variational analysis, as described by Ihde (2006), explores the multiple potential uses of a technology to better understand human-technology-world relations, emphasizing how technologies can be reimagined across different contexts. Artefacts, such as the notes produced during the reflective activity, served as an additional source of insight into learners' note-taking practices. These artefacts enabled exploration of postphenomenological concepts such as transparency and embodiment in relation to technology use. Furthermore, visual materials can reveal latent meanings that remain unspoken during interviews, complementing the manifest meanings associated with note-taking and its related technologies (Knowles & Sweetman, 2004). Visual methods contribute to the richness of qualitative data by uncovering additional layers of meaning, enhancing validity and depth, and supporting knowledge creation. They complement traditional data collection methods by capturing more nuanced and diverse forms of information than verbal or written responses alone. To analyze these visual materials, content analysis of images was conducted following principles of Visual Research Methodology (Rose, 2007).

In our data analyses we paid close attention to participants’ habits, relationships, routines, and the social factors influencing their note-taking practices. Ihde (2006) emphasizes the concept of variational multistability, suggesting that technologies can take on an indefinite range of stable states across cultural contexts. By reflecting on these variations within a form of technology, we gain deeper insights into its diverse applications and implications. Variational analysis, as described by Rosenberger (2014), incorporates both the user’s intentions and habitual preferences toward technology, as well as how others have materially modified or customized the same technology for their own purposes. This approach allowed us to explore not only the functionality of note-taking tools but also the nuanced ways these tools mediate ethical and academic decisions in educational settings.

4 Results

4.1 Learners’ familiarity and engagement with technology

Findings indicated that technology provides diverse ways for learners to engage with educational materials, but learners often lack awareness of these technologies and their potential implications for academic integrity. While participants frequently utilize various digital tools, they exhibit mixed feelings regarding the adoption of technology-driven note-taking methods, expressing excitement, confusion, and even resistance. The majority favoured familiar tools such as MS Word, Google Docs, and cloud-based platforms primarily due to convenience and ease of use (see Table 2). Participant 8 emphasized this preference by stating, “I would say use the tools that you are most comfortable with and that you find enjoyable to use.”

Responses varied considerably regarding technology integration into learning, confirming previous research (Mueller & Oppenheimer, 2014; Stacy & Cain, 2015). Concerns about academic integrity and ethical technology usage were prevalent among students, highlighting the need for clear guidance on what constitutes ethical

Table 2 AI-enhanced technology mentioned by participants (Original work Chudaeva (2024))

Technology tool	P1	P2	P3	P4	P5	P6	P7	P8
MS Word	v	v	v	v	v	v	v	v
MS Excel							v	
MS Office 365		v	v				v	
MS OneNote					v		v	
MS OneDrive								v
Google Sheets					v			
Google Slides			v					
iPad				v				
iBooks				v				
Smartphone (to access readings and discussion forums)		v		v				v
ChatGPT					v			
LMS (course content, discussion forums, dropbox)	v	v	v	v	v	v	v	v
Videoconferencing tools	v	v	v	v	v	v	v	v

Note: P2, for example, means ‘Participant 2’

note-taking (Kiewra et al., 2018; Morehead et al., 2019a, b). Concerns about authorship and intellectual property in educational contexts underscore the importance of instructional support (Luo et al., 2018).

Despite prevalent use of cloud-based tools, participants demonstrated minimal concern regarding privacy and data protection, prioritizing practicality. Behavioral observations, such as camera usage during remote learning, showed varying levels of engagement, potentially influencing note-taking practices (Castelli & Sarvary, 2021). Despite increased attention to AI tools like ChatGPT, only one participant reported using it as of February 2023. This participant appreciated its potential for synthesizing information and supporting brainstorming, describing it as a “dynamic literature compiler.” Participants consistently emphasized the necessity of clearer guidelines and explicit instruction on ethical technology usage.

As well, participants commonly took screenshots during online lectures without seeking explicit permission, viewing it as acceptable for personal study but differentiating this from formal academic submissions. *Participant 4* expressed this common perception, stating, “I don’t know what could ethically be wrong if I take a screenshot.” Additionally, learners commonly viewed copying and pasting as acceptable for personal notes but recognized the necessity of careful attribution in formal assignments. Participants prioritize convenience and effectiveness of learning strategies to assessing ethical impact of their actions, especially in learning activities perceived by learners as concerning only their actions (e.g., Williams, 2002). For example, Participant 1 said “*My time is limited so I need to make the most valuable use of it*” and Participant 5 explained their learning strategies as “*My decision is at the moment*”. Also, Participant 8 recommends “*I would say use the tools that you are most comfortable with and that you find enjoyable to you use.*” Participant 6 reflected on their choice of tools “*There are so many tools available and no one tool is the perfect tool for every context. So, I’ve just found the ways without even really thinking too critically about them [tools], just what tends to work best for me*”.

4.2 Image content analysis results

Figure 1 presents a comparative analysis of digital and analog notes. The results may be attributed to the different affordances of technologies used by participants. Some codes are more frequent features of paper images (e.g., ‘notes are short phrases’ and ‘some words are underlined’), some features are more commonly used in digital images (e.g., ‘table of content present’ and ‘notes are nice’), and some features of notes are used both in analog and digital notes (e.g., ‘Notes are words’ and ‘Main notes are black’).

5 Discussion

This section addresses the two guiding questions of the study, organized by learners’ familiarity and engagement with technology and their perspectives on academic integrity, especially as it relates to note-taking behaviors in educational contexts.

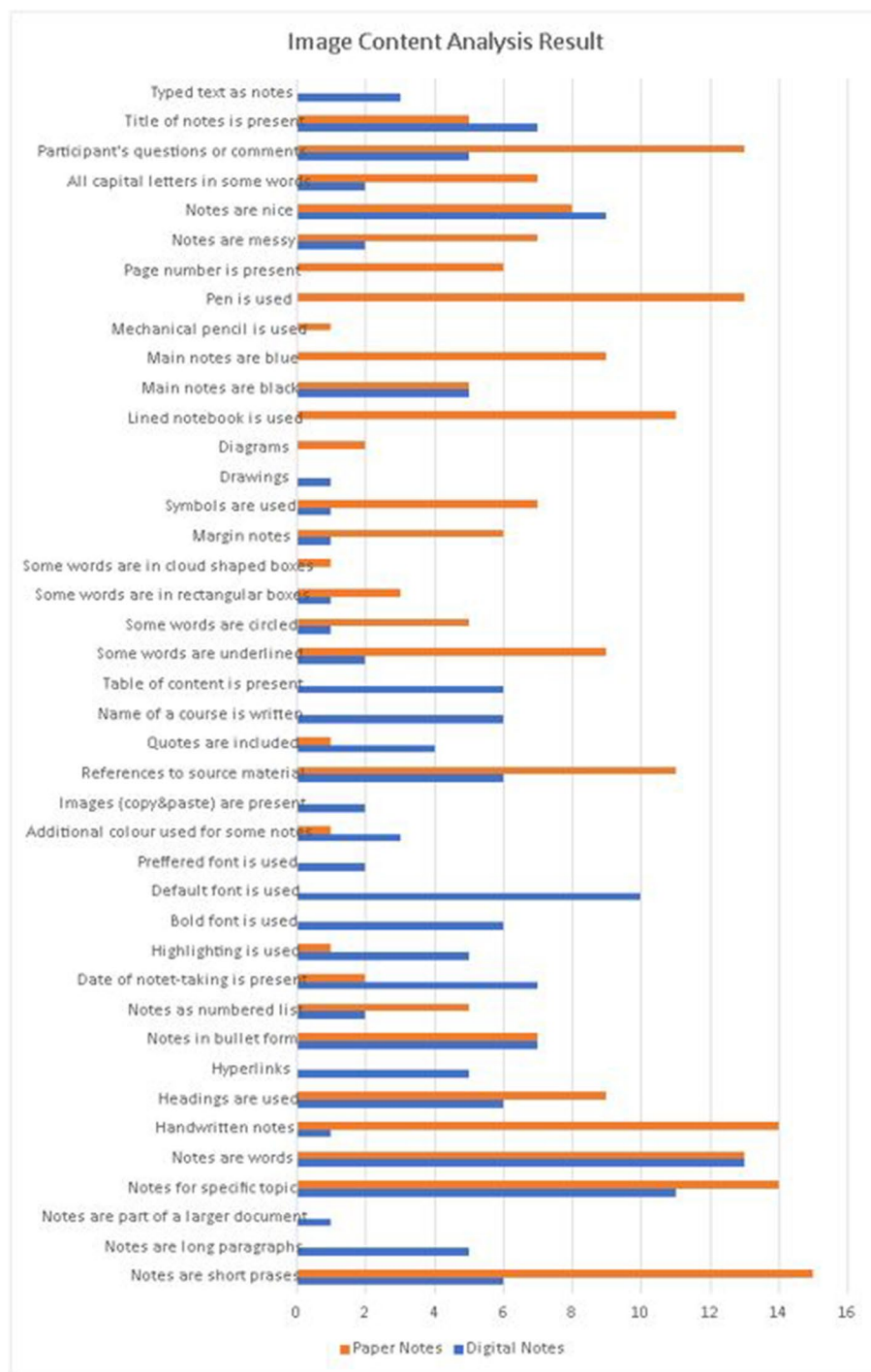


Fig. 1 Results of quantitative analysis of images

Technology continues to expand opportunities for engagement with educational materials and activities, offering learners multiple ways to interact with and organize information. However, consistent with previous research, learners often lack awareness of these new modes of interaction and the potential academic implications, highlighting the ongoing challenge of effectively guiding technology adoption in educational contexts (Mueller & Oppenheimer, 2014; Stacy & Cain, 2015). The increasing presence of AI in education provides opportunities for personalized learning, yet institutional guidance remains insufficient regarding ethical integration of AI-assisted work. Similar to findings by Hwang and Chen (2023), our study underscores the need for discussions and frameworks on responsible AI usage, including appropriate attribution and citation practices when AI contributes to academic tasks. Educating learners about responsible and ethical AI use is crucial to prevent misuse and promote positive learning practices (Zhai et al., 2024).

In exploring learners' approaches to note-taking with technology in academic contexts, several key themes have emerged that highlight both opportunities and challenges. Firstly, learners utilize a variety of digital tools to organize and review information, experimenting with different formats and personalized methods (Mueller & Oppenheimer, 2014). However, their responses to the integration of technology-driven tools for note-taking are mixed, ranging from excitement to confusion and resistance (Stacy & Cain, 2015). This variability suggests a need for clear guidance on how to effectively incorporate these technologies into learning practices. Concerns about academic integrity and the appropriate use of technology in note-taking are prevalent among higher education students (Kiewra et al., 2018). Learners express the need for explicit instructions on what constitutes ethical note-taking and the boundaries of acceptable technology use to uphold academic standards (Morehead et al., 2019a, b). The use of digital tools prompts questions about authorship and intellectual property in educational contexts, underscoring the importance of providing instructional support in these areas (Luo et al., 2018).

Behavioral observations in online environments have revealed varying levels of engagement, especially in relation to whether learners have cameras on or off during remote learning (Castelli & Sarvary, 2021). These behavioral differences likely influence note-taking habits and overall attitudes toward technology use in learning, suggesting that addressing these differences is essential for fostering responsible technology practices. The integration of technology in note-taking practices raises ethical concerns that necessitate the development of clearer guidelines in higher education institutions (Flanigan & Kiewra, 2018).

5.1 Question 1: what factors affect learners' decision-making and/or selection of technologies for note-taking?

Consistent with findings from previous research (Henderson et al., 2021), learners demonstrated minimal concern for privacy and data protection, prioritizing practicality and convenience, particularly when using cloud-based platforms. Furthermore, while some participants exhibited higher technological proficiency, the limited adoption of emerging tools like generative AI among learners in this study suggests

gaps in digital and AI literacy. This aligns with recent research by Hwang and Chen (2023), indicating the need for increased educational support for AI literacy.

5.1.1 Participant experiences with educational technologies

The study reveals that online learners engage with various technologies to meet their academic needs (see Table 2). Many tools, particularly cloud-based platforms, are favored due to their convenience in automatically backing up work. Despite the prevalent use of cloud-based tools, participants expressed minimal concern about privacy and data protection, emphasizing practicality over security. Furthermore, while some participants demonstrate more technological proficiency than others, emerging tools—particularly generative AI—are not widely adopted, suggesting a gap in digital and AI literacy.

5.1.2 AI and generative technology use

The limited adoption of ChatGPT among participants—only one out of eight used it as of February 2023—suggests that learners tend to rely on familiar tools. This preference underscores a broader need for AI literacy development at all educational stages, as learners' note-taking practices and technological use are often outpaced by advancements in tool development.

The study highlighted participants' varying levels of comfort with Artificial Intelligence (AI) and generative AI tools like ChatGPT (OpenAI, 2022). Although generative AI is not entirely new, its role in education remains ambiguous, with higher-education institutions still deliberating on policies and guidelines for its use. Remarkably, as interviews were conducted in early 2023—just three months after ChatGPT's public release on November 30, 2022 (OpenAI, 2022)—only one participant had experience using it.

This participant leveraged ChatGPT as a supplemental resource, using it to enhance understanding and gain additional perspectives on a topic, rather than as a primary source of answers. Mogavi et al. (2023) noted that the process of text and idea generation with AI raises concerns among educators, particularly since AI tends to oversimplify the information gathering and writing steps; however, this participant viewed ChatGPT as a supportive, brainstorming aid. The participant expressed appreciation for the language capabilities of the tool and also recognized its limitations, reflecting, *"I was interested to see what kind of synthesis or summary it might have and compared it with what I found when I'm going through my library databases."* This approach illustrates how generative AI tools can serve as flexible resources, allowing learners to access and review a range of information that can be explored in greater depth. In using ChatGPT, Participant 5 cited their process of saving and revisiting conversations and notes stored on an online server, treating the tool as a dynamic literature compiler that could assist with resource gathering, akin to a personal tutor (Hwang & Chen, 2023). This functionality supports Hwang and Chen's (2023) notion of AI as a personal guide, offering insights and organizing content in ways that enhance comprehension and streamline research efforts.

Participant 5 also share their ChatGPT experience as a feedback mechanism for academic writing. A recurrent theme among several participants mentioned a desire for more real-time interaction with peers and professors, a gap that AI technologies like ChatGPT could help fill (e.g., Zerkouk et al., 2024; Zerkouk et al., 2025). For instance, some learners discussed the possibility of using ChatGPT to refine iterative drafts, leveraging it as a peer-review assistant or a tool for verifying content reliability. Participant 5 observed that *“I can go back to this conversation, and I can look up the sources that ChatGPT offers.”* This comment underscores the participant’s view of ChatGPT as a reliable cross-reference that enables searchable and documented sources in one place.

Overall, these insights suggest that AI tools, particularly generative AI, can act as valuable supports in online learning environments, offering new avenues for interaction and enabling learners to engage in independent exploration. For distance learners, the ability to access immediate support and feedback from AI-powered tools like ChatGPT can be especially significant, serving as an accessible “conversation partner” that enhances the learning experience.

5.2 Question 2: what are learner’s perspectives on academic integrity when taking notes for learning?

Technologies, as Verbeek (2011) argues, can subtly influence behaviors and interactions, encouraging shifts in practices that may impact academic integrity. Our analysis reveals the nuances in learners’ ethical perspectives on note-taking, especially regarding actions like copying and pasting, taking screenshots, and referencing AI-generated content, which may be seen differently depending on the context. Specifically, we explored these questions: *Do learners view copying and pasting as plagiarism in personal notes? Are they guided on the responsible use of Generative AI tools, and is permission required for screenshots in presentations? Do learners’ note-taking etiquettes differ between in-person and online settings?* These questions frame how learners navigate ethical considerations in technology-enhanced learning environments.

Participant 3 described their experiences of adapting note-taking methods in different contexts, sharing, *“Recently I attended a conference, and I didn’t want to be perceived as rude having my laptop open the whole time. I did bring a physical notebook and a pen, and I took paper-based notes, which felt a bit foreign to me.”* This reflection reveals how in-person and online settings evoke different behaviors in learners. In an online environment, Participant 3 found it beneficial to take screenshots, stating, *“The other cool thing about distance learning ...is we’re online. So sometimes the professor would pull up a great slide, I just do a quick screen capture and dump that screen capture into my document with a little bit of explanation.”* The digital convenience of screenshotting becomes a normalized and efficient note-taking strategy, reinforcing Heidegger’s (1977) assertion that technology is not a mere means but “a way of revealing” learners’ ethical frameworks within their academic practices.

Most participants mentioned that they frequently take screenshots without seeking permission from presenters, assuming it is permissible if the content is shared

publicly. For instance, Participant 4 said, *“I think you could take screenshots of stuff. I don’t know what could ethically be wrong if I take a screenshot of something.”* This perspective highlights a distinction between personal note-taking and formal academic submissions; participants feel justified in capturing content for private use, whereas submitting such content in assignments would require more stringent adherence to academic integrity standards. Participant 7 expressed a similar sentiment about personal copying, saying, *“I would copy and paste, read aloud, or somehow transcribe interesting passages... [these] serve me well as reference materials for upcoming assignments.”* This casual approach to note-taking often shifts when learners prepare formal work, emphasizing a nuanced ethical line that separates personal study practices from academic submissions.

5.2.1 Complexities of copying and pasting in personal notes versus assignments

A recurring theme among participants was the dual perception of copying and pasting as both acceptable and unethical, depending on context. Participant 4 articulated this duality, noting, *“I might find some quotes I like and write things that I want on the side of the paper. I’ll put a little note saying how I should word it so that I don’t accidentally plagiarize.”* For many, the act of copying and pasting, while permissible for personal use, requires careful adjustment and attribution in formal assignments to avoid plagiarism. This divergence reflects how learners navigate a shifting ethical landscape, where technology facilitates convenient note-taking but also demands awareness of academic integrity standards. This finding also reminds of Ihde’s technological intentionality and mediated relations between humans and the world (Verbeek, 2001): learners’ experiences with taking a screenshot and the perceptions of such an action are directed by the convenience of the tools they use.

Our findings suggest that learners assume personal notes are private and, thus, less subject to academic scrutiny. Consequently, practices like copying, pasting, and screenshotting—typically considered academic misconduct in assignments—are often perceived as ethical in personal contexts. This distinction underscores how learners view technology-mediated note-taking as a flexible space for individual learning, free from the constraints of academic guidelines. However, when such notes become sources in formal assignments, learners may struggle to properly attribute the information, raising potential issues with plagiarism and intellectual property.

5.2.2 Impact of familiarity and accessibility on technology choices

Learners’ choices in note-taking technologies are also influenced by accessibility and familiarity. For example, some participants noted they felt overwhelmed by rapidly evolving digital tools, preferring to rely on familiar devices. Online learners often balance numerous technological tools, which can limit their willingness to adopt new ones solely for note-taking. This preference for familiar technology underscores the importance of institutional support for digital literacy, even for online learners who are often assumed to be technologically adept. One international learner participant shared that they only began online learning after gaining access to reliable Wi-Fi, while another learner living in a rural area opted to store notes locally due to limited internet

reliability. Such factors, as Rosenberger (2014) highlights, reflect how both material conditions and social contexts mediate learners' technology use. These infrastructure-related barriers also point to broader institutional responsibilities. For example, learners in rural or low-bandwidth environments may be disproportionately affected by AI tools that rely on high-speed connectivity or camera-on participation. In such cases, institutions must ensure accessibility by supporting offline or low-data alternatives and adopting anti-surveillance policies that respect learner privacy. Creating equitable learning environments means addressing not only technological proficiency, but also the infrastructural and ethical conditions that shape digital participation.

5.3 Ethical considerations and implications for academic integrity in digital Note-Taking

These findings raise important questions about how online learners use note-taking technologies and the implications for academic integrity. For instance, if learners take screenshots or copy text without tracking sources, they may later encounter difficulties when referencing these notes in assignments. This challenge points to broader issues around plagiarism, intellectual property, and authorship, particularly if learners fail to document original references or attribute content correctly. Many participants indicated that their initial intent in note-taking was personal use, leading to potential ethical blind spots when these notes later inform academic work. The analysis also reveals a gap in institutional guidance on ethical note-taking practices in online settings, as most participants reported no formal discussion on tools or ethical considerations around technology use. Academic conduct conversations typically focus on assessments, leaving out critical discussions on responsible note-taking. As technology continues to evolve, incorporating discussions on the ethical use of note-taking tools within academic conduct guidelines may become essential to foster integrity and responsible technology use among learners.

Rosenberger's (2014) questions about technological agency further illuminate these complexities, prompting reflection on what social and material factors enable specific uses of technology. This analysis of learners' practices reveals that while technology facilitates convenient access to information, it also challenges learners to navigate ethical boundaries, particularly in contexts where the distinction between personal and academic use is blurred. By examining how technology mediates note-taking behaviors, this study contributes to a deeper understanding of the relationship between academic integrity and digital learning practices in online education.

5.4 Implications of AI technologies on learner perspectives on academic integrity

The integration of artificial intelligence (AI) in note-taking technologies is transforming the ways learners in online education approach academic work and conceptualize academic integrity. These AI tools do not merely facilitate note-taking; they mediate and reshape learners' relationships with information, authorship, and ethical boundaries. AI-based note-taking functionalities offer significant benefits, such as enhanced efficiency and accessibility, which can improve academic performance. However, these tools also challenge traditional understandings of integrity and ethical learn-

ing practices, as they often blur the line between personal study aids and academic sources, especially when learners rely on AI-generated content for assignments.

In light of these shifts, educators play a crucial role in fostering an ethical awareness among learners about AI-enhanced note-taking practices. Based on our study findings within an online learning context, distance educators can proactively address the ethical complexities introduced by AI in note-taking technologies by:

- **Defining Academic Integrity and Misuse of AI:** Educators should clarify the concept of academic integrity, emphasizing specific misuses of AI tools that may constitute academic misconduct. For example, learners need to verify the AI-generated content to ensure it is representative, free of errors and accurate. This can help learners recognize when their use of AI crosses ethical lines.
- **Reviewing Institutional Policies on Technology Use:** At the beginning of each term and before major assessments, educators should review academic integrity policies with learners, including guidelines for responsible technology use. This reinforces the institution's expectations and promotes informed, ethical behavior.
- **Modelling Critical Engagement with AI:** By integrating AI tools into teaching practices and openly discussing AI outputs, educators can model a critical approach to AI-enhanced content. This includes examining AI's limitations, biases, and appropriate use, helping learners develop a nuanced understanding of these tools.
- **Educating Learners about Expectations for AI Citation Practices:** Learners should learn to cite AI tools, similar to other sources, and to clearly document the purpose and rationale for their use. This practice fosters transparency and responsibility in academic work.
- **Addressing Privacy and Security Concerns:** Educators should remind learners regularly about the privacy and security risks associated with AI tools, emphasizing the importance of responsible use and personal data protection.

By implementing these strategies, educators can encourage learners to critically reflect on their note-taking practices, helping them to navigate ethical challenges in a digitalized learning environment.

6 Conclusion

The rise of AI in educational tools calls for continued investigation into how AI literacy can be cultivated in higher education, particularly in ways that reshape learners' perceptions of academic integrity, authorship, and ownership of knowledge. As technology increasingly mediates learners' ethical considerations and academic identities, it influences how they understand and navigate the boundaries of academic conduct. This study has several limitations that should be considered when interpreting the findings. First, the relatively small sample size may limit the generalizability of the results to broader populations. Second, the potential for self-selection bias exists, as individuals who chose to participate may differ in meaningful ways from those who did not. Third, the rapid evolution of AI-powered technologies since the time of data collection may have altered how learners engage with these tools. The study was con-

ducted during the initial emergence of generative AI tools, which may explain why only one participant reported using ChatGPT, while others engaged with AI-powered tools embedded in learning management systems or productivity software such as the Microsoft Suite. Future research should include larger and more diverse samples and adopt longitudinal designs to explore the evolving impact of AI-enhanced note-taking on learners' ethical perspectives. Such studies could provide insight into how attitudes toward academic integrity and technology use shift over time, offering a deeper understanding of how AI not only transforms the tools of learning but also the values and norms that guide academic behavior.

Appendix 1 initial interview questions

Demographic Information.

1. What is your age range?
 - 20–29.
 - 30–39.
 - 40–49.
 - 50–59.
 - 60–69.
2. What is your gender? (to identify any gender differences)
 - Female.
 - Male.
 - Non-binary/third gender.
 - Prefer to self-describe.
 - Transgender.
 - Cisgender.
 - Agender.
 - Genderqueer.
 - A gender not listed.
 - Prefer not to answer.
3. How many year(s) of Study in Post-Secondary Education do you have?
4. How many years of study in an online program do you have?

Conceptualization.

1. Think of a time when you took notes for learning in your program and describe that in as much detail as possible.
2. Tell me about the ways you take notes for learning.
3. Tell me about your typical way to study in the program.
4. How would you describe the experience of taking-notes?

Apprehending the phenomenon.

1. Tell me about what you do when you decide to make notes.
2. Tell me how you decide what tool to use to take notes.
3. Tell me what you do with the notes.
4. If you wanted to teach me how to take notes, what would you tell me?
5. What, if any, are ethical questions that maybe related to your note-taking process?

Questions adapted from.

- Roulston, K. (2010). *Reflective interviewing*. Sage.
- Bevan, M., T. (2014). A method of phenomenological interviewing. *Qualitative Health Research*, 24(1), 136–144. <https://doi.org/10.1177/1049732313519710>.

Appendix 2 reflective activity questions

Look back at your notes.

- What do they look like?
- Are they good notes? How do you know?
- What will you do with your notes afterwards?
- How do you feel about learning by taking notes?
- What was hard or easy about taking notes?
- Are you comfortable taking notes by hand? Why?
- What have you learned about the way you learn by taking notes?
- What is note-taking?

Data availability The data supporting the findings of this study are available upon reasonable request from the corresponding author due to ethical and privacy restrictions. Due to the sensitive nature of the data, access may be granted upon request, subject to ethical approval and participant confidentiality agreements. No proprietary or third-party datasets were used in this study.

Declarations

Competing Interests The authors declare that they have no competing interests (e.g., financial, professional, or personal) with the editors or funding agencies that could have influenced the research within the last three years from the commencement of this work. Additionally, no part of this manuscript has been previously published, nor is it under consideration for publication elsewhere.

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